

The Program in Action

Figure 36-1 shows what the output will look like when you run *hourglass.py*.

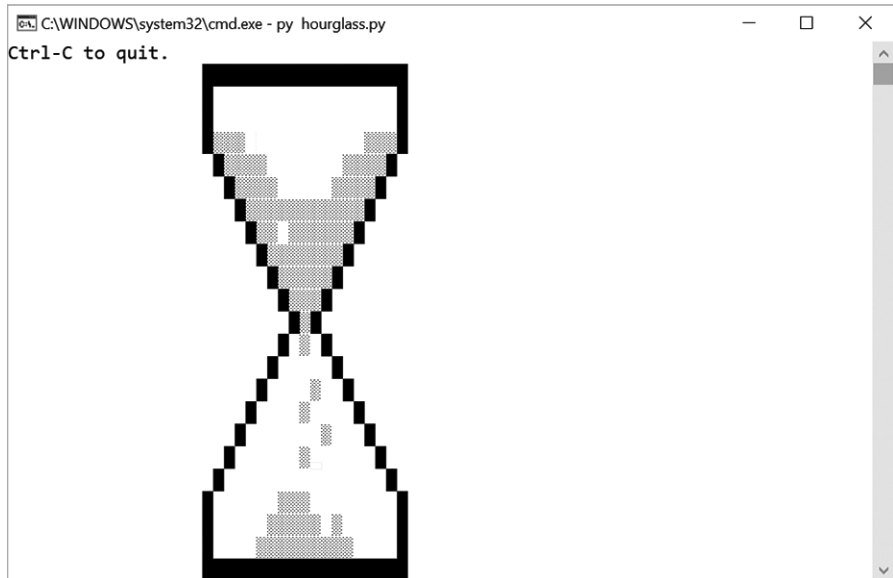


Figure 36-1: The output of the *hourglass* program with sand falling

How It Works

The *hourglass* program implements a rudimentary physics engine. A *physics engine* is software that simulates physical objects falling under gravity, colliding with each other, and moving according to the laws of physics. You'll find physics engines used in video games, computer animation, and scientific simulations. On lines 91 to 102, each “grain” of sand checks if the space beneath it is empty and moves down if it is. Otherwise, it checks if it can move down and to the left (lines 104 to 112) or down and to the right (lines 114 to 122). Of course, there is much more to *kinematics*, the branch of classical physics that deals with the motion of macroscopic objects, than this. However, you don't need a degree in physics to make a primitive simulation of sand in an hourglass that is enjoyable to look at.

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1. `"""Hourglass, by Al Sweigart al@inventwithpython.com`
 2. `An animation of an hourglass with falling sand. Press Ctrl-C to stop.`
 3. `View this code at https://nostarch.com/big-book-small-python-projects`
 4. `Tags: large, artistic, bext, simulation"""`
 5.
 6. `import random, sys, time`
 7.
 8. `try:`
 9. `import bext`
 10. `except ImportError:`
 11. `print('This program requires the bext module, which you')`